

Arshia Pessaran

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EDUCATION

Georgia Institute of Technology – MS in Computer Science (ML specialization)
University of Central Florida – BS in Data Science

expected Jun 2027
Dec 2023

PROFESSIONAL EXPERIENCE

Software Developer | *Seminole County Public Schools*

Jan 2024 – Present

- Wrote and maintained code for ASP.NET-based web applications, while managing code quality through peer reviews, branch approvals, and documentation initiatives. Provided ongoing production support through Kace ticket resolution and software bug fixes.
- Executed complex SQL queries for data-driven initiatives, including processing AP/IB test scores for 50,000+ teachers' performance bonuses and generating critical student threat assessment reports
- Developed ETL pipelines for state test file processing and transportation log integration with fault-tolerant workflows, logging, and email alerts

Key Projects:

- **Emergency Work Order System:** Architected full-stack solution for Facilities department, including database design, user-based security, and responsive UI with dynamic dashboard. Integrated Twilio SMS for real-time mechanic alerts, processing 30+ daily work orders. Led full project lifecycle from requirements to production.
- **CI/CD Pipeline Implementation:** Led CI/CD adoption, defining Git branching strategy and deployment protocols. Automated production pushes with Azure DevOps and YAML-based pipelines, replacing manual deployments with standardized, reliable workflows.
- **Teacher & Administrator Evaluation Model:** Designed VAM in R using logistic regression and weighted t-tests to assess educator performance across 34% of annual reviews. Incorporated student growth, class size, and demographics while ensuring Florida DOE compliance. Model informs district-wide compensation and retention decisions.
- **Automation Framework Migration:** Replaced costly Katalon Studio macros with C# Selenium WebDriver, then modernized automation with Playwright. Eliminated \$15K in annual licensing costs while improving test reliability, execution speed, and maintainability.

Statistical Programmer Intern | *Intego Group*

May 2023 – Aug 2023

- Processed clinical trial data using SAS to create SDTM and ADaM datasets, implementing CDISC standards and WHODrug dictionaries for regulatory compliance
- Generated statistical analyses and data visualizations from ADaM datasets to support clinical study reports and drug development presentations

PROJECTS

Nautiq – Official Deep6 Dive App | *Swift, SwiftUI, Kotlin, Jetpack Compose, CoreBluetooth*

Jan 2026 – Mar 2026

- Reverse-engineered a proprietary BLE protocol through HCI snoop log captures, Android APK decompilation, and live hardware testing to implement dive sync, device settings, and over-the-air firmware updates beyond what any existing app or open-source library provided
- Implemented the first iOS firmware update path for the Deep6 Excursion dive computer via an undocumented FFE2 channel activated by a magic trigger sequence, closing a gap the manufacturer had left open since 2022
- Shipped to TestFlight and pitched directly to the manufacturer, securing official endorsement as their companion app with full branding and firmware collaboration rights
- Built native iOS (SwiftUI/SwiftData/CloudKit) and Android (Jetpack Compose/Room) apps with zero third-party dependencies, verified against 77 dives from a physical device

Bludly Health Analytics Platform | *React, Node.js, Express, PostgreSQL, Docker, AWS*

Jan 2025 – Present

- Designed complete system architecture for healthcare analytics platform, making critical infrastructure decisions around data flow, security model, and scalability planning while maintaining strict HIPAA compliance
- Developed responsive React front-end with interactive D3.js visualizations, multi-profile management, and intuitive lab data handling, while implementing secure Node.js/Express back-end with a Dockerized SQL database, RESTful APIs, and a multi-tiered subscription system
- Engineered AI-powered clinical analysis system using self-hosted Deepseek model, designing custom OCR document processing pipeline, spaCy-based NLP for medical data extraction, and cost-optimized deployment strategies

SKILLS

Software Development: ASP.NET MVC, C#, JavaScript (jQuery/AJAX), HTML/CSS, Azure, CI/CD, Git

Mobile & Systems: Swift/SwiftUI, Kotlin/Jetpack Compose, CoreBluetooth, BLE Protocol Engineering

AI & Data: Python, Airflow/ETL, SQL, R, PyTorch, NLP, LangChain, Statistical Modeling, pandas/NumPy, Jupyter

Professional Competencies: Full-Stack Development, AI, Database Development, Cloud Solutions